BANKING ANALYTICS PROJECT

Model Design Document (MDD)

Insights for Smarter Financial Decisions

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Database: Banking_Analytics (Table: banking_data)
Tools: Excel · Python · SQL · Power BI

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1. Project Overview

The Banking Customer Analytics Project analyzes customer demographic and financial data to uncover insights that support data-driven decision-making in a retail banking context. The analysis covers customer profiles, banking behavior, deposits, loans, credit exposure, and risk classification, and is delivered through three layers:

- Python (EDA) — data cleaning, validation, and exploratory analysis (Banking_Analytics.ipynb).
- SQL (Business Analysis) — structured querying of the cleaned dataset across 8 scripts, from basic filtering to views/CTEs.
- Power BI (Dashboard) — a 5-page interactive dashboard, "Banking Analytics — Insights for Smarter Financial Decisions" (Banking_Analytics_Dashboard.pbix).

2. Business Problem

The bank wants to understand customer demographics, financial behavior, and product usage to support data-driven decision making. The analysis targets:

- Identifying valuable and high-potential customer segments
- Understanding financial behavior (deposits, loans, savings, credit usage)
- Assessing customer risk exposure
- Supporting retention, cross-sell, and wealth-management strategy

3. Dataset Summary

Attribute	Value
Records	3,000 customers
Features	25 columns
Grain	One row per unique customer
Format	Structured CSV → SQL table → Power BI model
Missing values	0 (fully clean)
Duplicate Client IDs	0

4. Data Dictionary

Raw Column	SQL Column	Type	Description
Client ID	client_id	Text	Unique customer identifier
Name	name	Text	Customer full name
Age	age	Number	Customer age
Location ID	loction_id	Number	Branch/region location code
Joined Bank	joined_bank	Date	Account opening date
Banking Contact	banking_contact	Text	Assigned relationship manager

Raw Column	SQL Column	Type	Description
Nationality	nationality	Text	Customer nationality group
Occupation	occupation	Text	Customer profession
Fee Structure	fee_structure	Text	Fee tier (High / Mid / Low)
Loyalty Classification	loyalty_classification	Text	Tier: Jade, Silver, Gold, Platinum
Estimated Income	estimated_income	Number	Estimated annual income
Superannuation Savings	superannuation_savings	Number	Retirement savings balance
Amount of Credit Cards	amount_of_credit_cards	Number	Number of credit cards held
Credit Card Balance	credit_card_balance	Number	Outstanding credit card balance
Bank Loans	bank_loans	Number	Total loan balance
Bank Deposits	bank_deposits	Number	Total deposit balance
Checking Accounts	checking_accounts	Number	Checking account balance
Saving Accounts	saving_accounts	Number	Savings account balance
Foreign Currency Account	foreign_currency_account	Number	Foreign currency balance
Business Lending	business_lending	Number	Business loan exposure
Properties Owned	properties_owned	Number	Count of owned properties
Risk Weighting	risk_weighting	Number	Risk score (1 = lowest, 5 = highest)
BRId	br_id	Number	Branch reference ID
GenderId	gender_id	Number	Gender code (1 / 2)
IAId	ia_id	Number	Investment advisor reference ID

5. Project Architecture / Workflow

Banking.csv (raw source, 3,000 rows × 25 columns)

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Python EDA (Banking_Analytics.ipynb) — load & inspect, missing value / duplicate checks, date conversion, univariate and bivariate visual exploration.

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MySQL Database (Banking_Analytics.banking_data) — table creation, standardized column names, and 8 layered SQL scripts from basic retrieval through views/CTEs.

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Power BI (Banking_Analytics_Dashboard.pbix) — a 5-page interactive dashboard for stakeholder reporting.

6. SQL Analysis Layer Summary

Script	Purpose
01_Database_Setup.sql	Creates the Banking_Analytics database and standardizes all 25 column names to snake_case.
02_Basic_SQL.sql	Core retrieval queries: filtering by age/income/deposits/risk, LIKE/IN filters, top-N sorting.
03_Aggregate_Functions.sql	Bank-wide KPIs: total customers, average income/deposits/loans, min/max ranges, totals.
04_Group_By_Having.sql	Segment-level metrics by nationality, occupation, loyalty tier, gender, and risk weighting, filtered with HAVING.
05_Subqueries.sql	Customers/segments compared against overall averages (above/below-average income, deposits, loans, credit balance).
06_Window_Functions.sql	Reserved for ranking/running-total analysis (not yet implemented).
07_Banking_Business_Analysis.sql	Applied business logic: Premium Banking eligibility, Wealth Management eligibility, high-risk/high-loan exposure, investment opportunities, financial relationship value ranking.
08_Views_and_CTE.sql	Reusable views: premium_customers, high_risk_customers, investment_opportunity, top_depositors.

7. Power BI Data Model

- Fact table: banking_data — one row per customer, containing all demographic and financial attributes.
- Grain: Customer-level (3,000 rows).
- Key measures (cards): Total Deposits, Total Loans, Total Savings, Average Income, Average Risk, Total Customers, High-Risk Customer %, Average Loan (High Risk), Highest Customer Deposit, Highest Customer Loan.
- Slicers used across pages: Gender, Nationality, Loyalty Classification, Risk Category, Fee Structure, Income Category, Age Group.
- Navigation: Each page includes a left-hand navigation rail with icons for all 5 pages, plus a page-reset control top-right.

8. Dashboard Pages

The Power BI report (Banking_Analytics_Dashboard.pbix) contains 5 pages, each carrying the "Bank Analytics Dashboard" branding, a left navigation rail, top KPI cards, and slicer filters. Figures below are read directly from the published report.

8.1 Page 1 — Executive Overview

A high-level snapshot of the entire customer book for leadership review.

Figure 1 — Executive Overview page

KPI Cards

Total Customers	Total Deposits	Total Loans
3K	**2.0bn**	**1.8bn**

Total Savings	Average Income	Average Risk
698.7M	**171.3K**	**2.25**

Slicers

- Gender · Nationality · Loyalty Classification · Risk Category

Visuals

Visual	Type	Key Figures
Customer Demographics	Treemap	Female: 1.51K · Male: 1.49K
Customer Age Distribution	Clustered Column Chart	17-25: 366 · 26-35: 457 · 36-45: 427 · 46-55: 459 · 56-65: 426 · 66+: 865

Visual	Type	Key Figures
Risk Profile	Donut Chart	Medium Risk: 41% · High Risk: 31% · Low Risk: 28%
Top 10 Occupations by Total Deposits	Clustered Bar Chart	Structural Analysis Engineer: 23M · Database Administrator III: 21M · Social Worker: 20M · Office Assistant III: 19M · Systems Administrator II: 18M · Software Engineer IV: 18M
Customer Distribution by Nationality	Donut Chart	European: 44% · Asian: 25% · American: 17% · Australian: 8% · African: 6%

8.2 Page 2 — Customer Analysis

A deep dive into customer demographics and segmentation behavior.

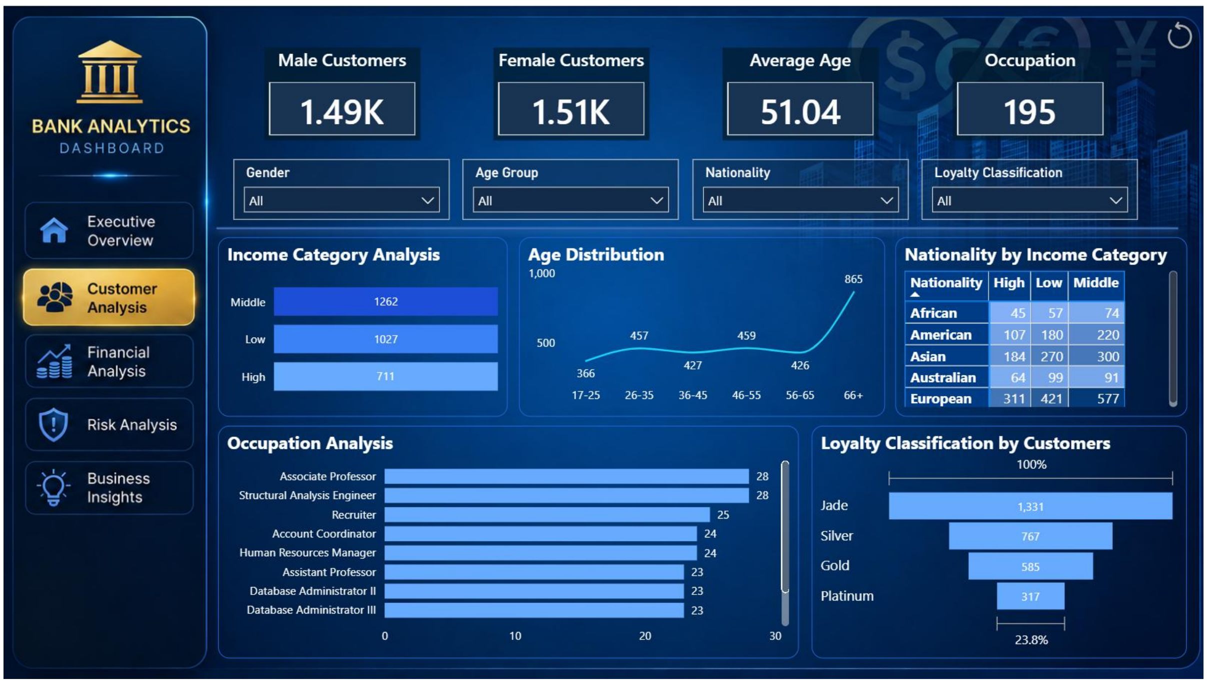

Figure 2 — Customer Analysis page

KPI Cards

Male Customers	Female Customers	Average Age	Occupation (unique)
1.49K	1.51K	51.04	195

Slicers

- Gender · Age Group · Nationality · Loyalty Classification

Visuals

Visual	Type	Key Figures
Income Category Analysis	Bar Chart	Middle: 1,262 · Low: 1,027 · High: 711 customers

Visual	Type	Key Figures
Age Distribution	Line Chart	17-25: 366 · 26-35: 457 · 36-45: 427 · 46-55: 459 · 56-65: 426 · 66+: 865
Occupation Analysis	Clustered Bar Chart	Associate Professor: 28 · Structural Analysis Engineer: 28 · Recruiter: 25 · Account Coordinator: 24 · HR Manager: 24 · Assistant Professor: 23 · Database Administrator II/III: 23
Nationality by Income Category	Pivot Table (Matrix)	European: High 311 / Low 421 / Middle 577 · Asian: 184 / 270 / 300 · American: 107 / 180 / 220 · Australian: 64 / 99 / 91 · African: 45 / 57 / 74
Loyalty Classification by Customers	Funnel Chart	Jade: 1,331 · Silver: 767 · Gold: 585 · Platinum: 317 (23.8% conversion Jade→Platinum)

8.3 Page 3 — Financial Analysis

Focuses on the bank's financial exposure — deposits, loans, savings, and fees.

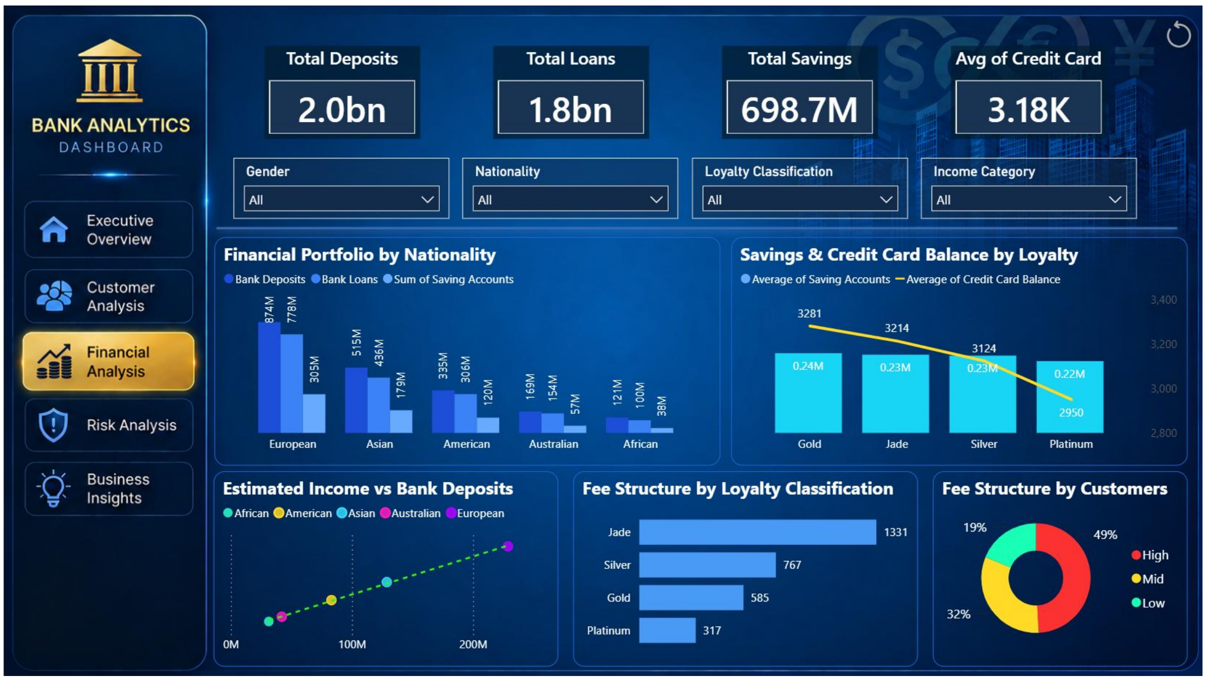

Figure 3 — Financial Analysis page

KPI Cards

Total Deposits	Total Loans	Total Savings	Avg. Credit Card Balance
2.0bn	**1.8bn**	**698.7M**	**3.18K**

Slicers

- Gender · Nationality · Loyalty Classification · Income Category

Visuals

Visual	Type	Key Figures
Financial Portfolio by Nationality	Clustered Column Chart	European: Deposits 874M / Loans 778M / Savings 305M · Asian: 515M / 436M / 179M · American: 335M / 306M / 120M · Australian: 169M / 154M / 57M · African: 121M / 100M / 38M
Estimated Income vs Bank Deposits	Scatter Chart	Positive trend across African, American, Asian, Australian, European segments — deposits rise with income
Savings & Credit Card Balance by Loyalty	Line + Clustered Column Combo	Avg. Saving Accounts — Gold: 0.24M · Jade: 0.23M · Silver: 0.23M · Platinum: 0.22M; Avg. Credit Card Balance line — Gold: 3,281 · Jade: 3,214 · Silver: 3,124 · Platinum: 2,950
Fee Structure by Loyalty Classification	Bar Chart	Jade: 1,331 · Silver: 767 · Gold: 585 · Platinum: 317
Fee Structure by Customers	Donut Chart	High: 49% · Mid: 32% · Low: 19%

8.4 Page 4 — Risk Analysis

Assesses customer credit and portfolio risk exposure.

Figure 4 — Risk Analysis page

KPI Cards

Average Risk	High-Risk Customers	Avg. Loan (High Risk)	High-Risk Customer %
2.25	942	839.9K	31%

Slicers

- Gender · Risk Category · Loyalty Classification · Fee Structure

Visuals

Visual	Type	Key Figures
Average Loan by Risk	Column Chart	High Risk: 840K · Medium Risk: 576K · Low Risk: 334K
Income vs Risk Category	Scatter Chart	Low Risk clusters near ~₹1,00,000K income · Medium Risk ~₹2,00,000K · High Risk extends past ~₹2,50,000K, showing risk is not purely income-driven
Avg Bank Deposits by Risk	Line Chart	High Risk: 954K · Medium Risk: 654K · Low Risk: 378K
Avg Credit Card Balance by Risk	Column Chart	High Risk: 4.45K · Medium Risk: 3.08K · Low Risk: 1.88K
Customer Risk Segmentation	100% Stacked Column Chart	By loyalty tier — Jade: Medium 42% / Low 27% / High 31% · Silver: 39% / 29% / 32% · Gold: 41% / 28% / 32% · Platinum: 38% / 30% / 32%
Risk Distribution	Donut Chart	Medium Risk: 41% · High Risk: 31% · Low Risk: 28%

8.5 Page 5 — Business Insights

Actionable, ranked views plus narrative insights and recommendations for relationship managers and strategy teams.

Figure 5 — Business Insights page

KPI Cards

Highest Customer Deposit	Highest Customer Loan	Total Customers	Unique Occupations
3.89M	2.67M	3K	195

Top 10 Depositors (sample rows shown)

Client ID	Name	Occupation	Nationality
IND68515	Ronald Reed	Safety Technician II	Asian
IND81510	Harry Burns	VP Quality Control	European
IND70565	Craig Hansen	Research Assistant II	European
IND91720	Harry Gibson	Software Engineer IV	European
IND47511	Johnny Patterson	Human Resources Assistant II	American
IND53236	John Frazier	Systems Administrator II	European
IND52376	Scott Hart	Safety Technician III	European

Top 10 Loan Customers (sample rows shown)

Client ID	Name	Occupation	Risk Category
IND23172	Ryan Mitchell	Programmer Analyst I	High Risk
IND22626	Terry Chavez	Geologist IV	High Risk
IND99978	Russell Wagner	Operator	High Risk
IND99607	Sean Pierce	Health Coach II	High Risk
IND54314	Kevin Gomez	Project Manager	High Risk
IND89040	Earl Hart	Database Administrator I	High Risk
IND85958	Gary Hunt	Research Associate	Medium Risk

Top 10 Occupation Financial Performance (sample rows shown)

Occupation	Sum of Bank Deposits
Structural Analysis Engineer	2,28,39,335.09
Database Administrator III	2,06,18,343.75
Social Worker	1,99,63,970.99
Office Assistant III	1,93,63,109.47
Systems Administrator II	1,79,94,545.02
Software Engineer IV	1,79,63,983.62
Software Consultant	1,69,76,075.32
Total (Top 10)	18,57,01,434.67

Key Business Insights (as reported on the dashboard)

- European customers contribute the highest deposit volume.
- Structural Analysis Engineer is among the strongest financial-performing occupations.
- High-value deposit customers represent important retention opportunities.
- Large loan customers should be evaluated alongside their risk category.
- Customer income and deposits show a meaningful relationship.

Management Recommendations (as reported on the dashboard)

- Retain high-value deposit customers through targeted relationship strategies.
- Closely monitor high-loan customers with High Risk classification.
- Develop targeted financial products for valuable customer segments.
- Use occupation and financial behavior to identify cross-selling opportunities.
- Combine risk and financial exposure when making lending decisions.

9. Key Business Outputs (from SQL Layer)

- Premium Banking candidates: above-average income and above-average deposits.
- Wealth Management candidates: above-average income & deposits and Gold/Platinum loyalty tier.
- Investment opportunity segment: above-average income but below-average deposits (under-penetrated wallet).
- High-risk exposure segment: risk weighting 4–5 with above-average bank loans.
- Financially strong segment: above-average income & deposits with low risk weighting (1–2).
- Top relationship value customers: ranked by bank_deposits + saving_accounts + credit_card_balance.

10. Reusable SQL Views

View	Definition
premium_customers	Income and deposits both above the bank-wide average
high_risk_customers	Risk weighting of 4 or 5
investment_opportunity	Above-average income but below-average deposits
top_depositors	Deposits above the bank-wide average

11. Cross-Validation: Dashboard vs. SQL / EDA Layer

The published dashboard figures align closely with the underlying Python EDA and SQL layers, confirming consistency across the pipeline:

- Total customers: 3,000 (EDA sheet) = 3K (dashboard, all 5 pages).
- Gender split: 1,512 Gender Code 2 vs. 1,488 Gender Code 1 (EDA) ≈ 1.51K Female vs. 1.49K Male (dashboard).
- Average age: 51.04 (EDA) = 51.04 (Customer Analysis page).
- Total bank deposits: ₹2,014,680,582 (EDA) ≈ 2.0bn (dashboard).
- Total bank loans: ₹1,774,158,466 (EDA) ≈ 1.8bn (dashboard).
- Total savings account balance: ₹698,725,060 (EDA) = 698.7M (dashboard).
- Top occupation deposit totals (e.g., Structural Analysis Engineer, Database Administrator III, Social Worker) match between the SQL Top-10-occupation query and the Business Insights page.

12. Future Enhancements

- Complete 06_Window_Functions.sql (ranking, running totals, moving averages by segment).
- Add predictive modeling (deposit/churn prediction) as referenced in the notebook title.
- Extend the Power BI model with a proper star schema (separate dimension tables for Branch, Gender, Investment Advisor) rather than a single flat fact table.
- Add drill-through from Business Insights back to Customer/Financial/Risk pages for a selected client.